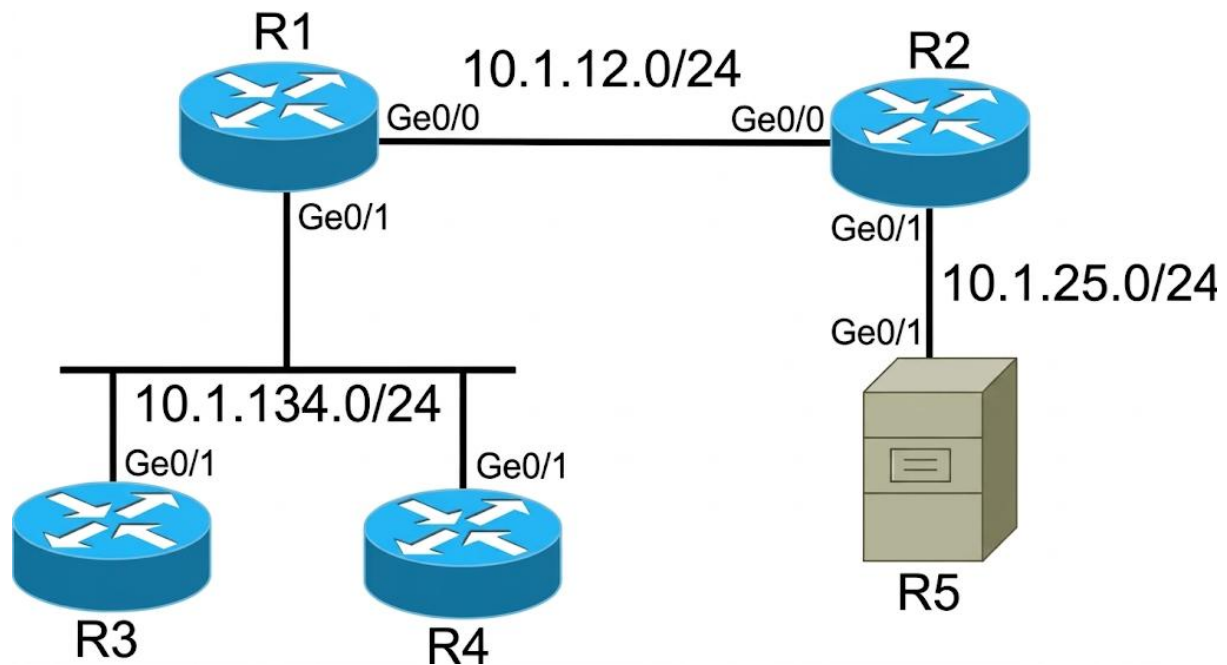


Named ACL

Technical Lab Guide & Configuration Documentation



Lab Objectives

1. **Host and IP Configuration:** Configure initial hostnames and IP interfaces across all devices as detailed in the network design.
2. **Dynamic Routing:** Configure OSPF Process 100 on all routers to advertise connected networks within Area 0.
3. **End-Host Emulation (R5):**
 - Disable the IP routing feature on R5 to act as an end-device.
 - Set the default-gateway to point to R2's `GigabitEthernet0/1` interface.
 - Enable the HTTP service on R5.
4. **Named Extended ACL Filtering (R1):**
 - Create an Extended Named ACL (`FILTER_HTTP`) applied inbound on `GigabitEthernet0/1`.
 - Explicitly deny HTTP (TCP port 80) traffic originating from R3 targeting R5.
 - Permit all other IP traffic.
5. **Named Standard ACL Filtering (R2):**
 - Create a Standard Named ACL (`FILTER_HOST`) applied outbound on `GigabitEthernet0/1`.
 - Explicitly deny R4 from reaching R5.
 - Permit all remaining hosts to access R5.

Addressing & Interface Plan

Device	Interface	IP Address
R1	GigabitEthernet0/0	10.1.12.1
R1	GigabitEthernet0/1	10.1.134.1
R2	GigabitEthernet0/0	10.1.12.2
R2	GigabitEthernet0/1	10.1.25.2
R3	GigabitEthernet0/1	10.1.134.3
R4	GigabitEthernet0/1	10.1.134.4
R5	GigabitEthernet0/1	10.1.25.5

Device Configurations

Router R1

enable

configure terminal

hostname R1

!

interface GigabitEthernet0/0

ip address 10.1.12.1 255.255.255.0

no shutdown

!

interface GigabitEthernet0/1

ip address 10.1.134.1 255.255.255.0

ip access-group FILTER_HTTP in

no shutdown

!

router ospf 100

network 10.1.12.0 0.0.0.255 area 0

network 10.1.134.0 0.0.0.255 area 0

!

ip access-list extended FILTER_HTTP

deny tcp host 10.1.134.3 host 10.1.25.5 eq www

permit ip any any

!

end

Router R2

t

enable

configure terminal

hostname R2

!

interface GigabitEthernet0/0

ip address 10.1.12.2 255.255.255.0

no shutdown

!

interface GigabitEthernet0/1

ip address 10.1.25.2 255.255.255.0

ip access-group FILTER_HOST out

no shutdown

!

router ospf 100

network 10.1.12.0 0.0.0.255 area 0

network 10.1.25.0 0.0.0.255 area 0

!

ip access-list standard FILTER_HOST

deny host 10.1.134.4

permit any

!

end

Router R3

enable

configure terminal

hostname R3

!

interface GigabitEthernet0/1

ip address 10.1.134.3 255.255.255.0

no shutdown

!

router ospf 100

network 10.1.134.0 0.0.0.255 area 0

!

end

Router R4

enable

configure terminal

hostname R4

!

interface GigabitEthernet0/1

ip address 10.1.134.4 255.255.255.0

no shutdown

!

router ospf 100

network 10.1.134.0 0.0.0.255 area 0

```
!
```

```
end
```

Router R5 (Configured as Host Server)

```
enable
```

```
configure terminal
```

```
hostname R5
```

```
no ip routing
```

```
ip http server
```

```
ip default-gateway 10.1.25.2
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.25.5 255.255.255.0
```

```
no shutdown
```

```
!
```

```
end
```

Verification

In this task, we have configured two access-lists on R1 and R2. The first access-list "FILTER_HTTP" is specifically defined to fulfill first task requirement, which filters http traffic initiated from the R3. The access-list "FILTER_HOST" is configured to deny R4 from reaching R5, which is applied outbound at GigabitEthernet0/1 interface on R2.

```
R3#telnet 10.1.25.5 80
```

```
Trying 10.1.25.5, 80 ...
```

```
% Destination unreachable; gateway or host down
```

```
!
```

```
R1#show access-lists
```

```
Extended IP access list FILTER_HTTP
```

```
10 deny tcp host 10.1.134.3 host 10.1.25.5 eq www (1 match)
```

```
20 permit ip any any (426 matches)
```

```
!
```

```
R3#ping 10.1.25.5
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.1.25.5, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 84/116/220 ms
```

```
⌘
```

In the above output, HTTP traffic is getting dropped on R1 due to the defined access-list and we can see traffic match in the access-list.

```
⌘R4#ping 10.1.25.5
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.1.25.5, timeout is 2 seconds: U.U.U
```

```
Success rate is 0 percent (0/5)
```

```
!
```

```
R2#show access-lists
```

```
Standard IP access list FILTER_HOST
```

```
10 deny 10.1.134.4 (8 matches)
```

```
20 permit any (5 matches)
```

```
!
```

```
R4#telnet 10.1.25.5 80
```

```
Trying 10.1.25.5, 80 ...
```

```
% Destination unreachable; gateway or host down
```

```
⌘
```

All the traffic from R4 to R5 is being filtered by ACL configured on R2 as expected.